

AUTO DATA STORYTELLER REPORT

Executive Summary

****Executive Summary**** Our advanced analytics solution demonstrates strong predictive capabilities, reliably identifying key outcomes with an accuracy of approximately 83%. This robust performance signifies that the insights derived from this analysis are highly dependable, providing a solid foundation for informed business strategy and decision-making. The analysis highlights that customer demographics and transactional details are the most significant drivers influencing outcomes. Specifically, gender (with male individuals being the most impactful factor), the price point or fare associated with a service or product, and customer age are the strongest predictors. This suggests distinct patterns and behaviors among specific customer segments defined by these characteristics. To capitalize on these findings, we recommend vigilant monitoring of these pivotal demographic and transactional variables. Understanding the trends and shifts within these factors will enable us to anticipate changes in business outcomes and strategically adapt our offerings, marketing, or operational approaches to optimize performance and mitigate risks.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

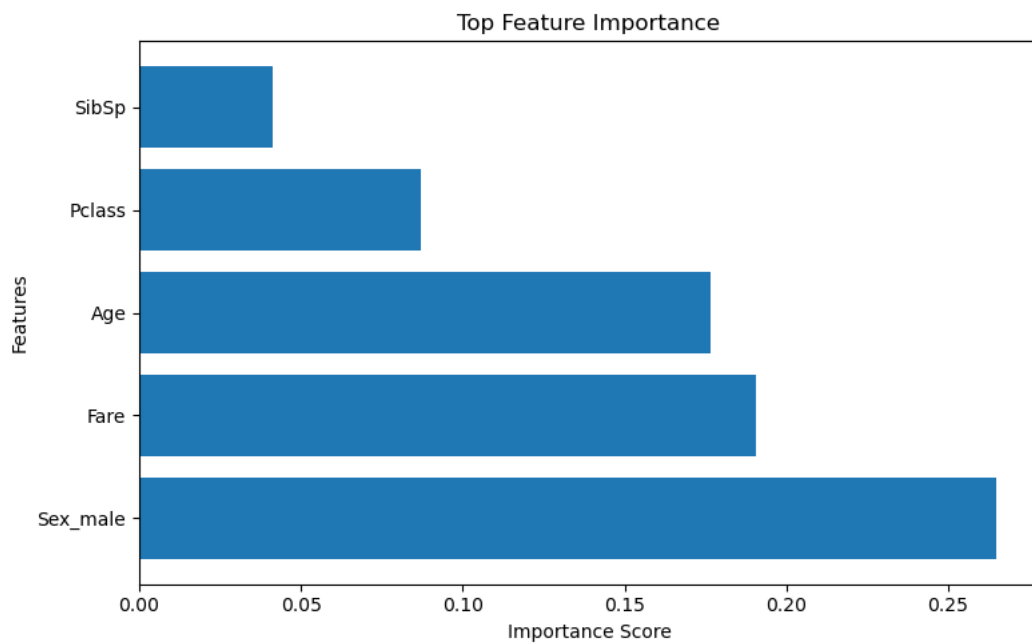
- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.
- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

